THE UNIVERSITY OF
NOTRE DAME
AUSTRALIA

MED 4000
WEEK 18

TUTOR GUIDE

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RELEVANT PRIOR LEARNING

MED 1000

1.01 Paediatric history
1.14 Immune system
1.26 Principles of paediatric examination

MED 2000

2.02 Microbiology of common skin infections
2.07 Assessment of the febrile child
2.22 Assessment and management of the febrile child
2.24 Assessment and management of bony lumps
2.25 Assessment of the swollen knee

MED 3000

3.28 Purpuric rashes in children
Acute leukaemia

WEEK 18_PAEDIATRICS MEDICINE		
YR	WK	LEARNING OBJECTIVES
4	18	• Know the key components of the history and examination in the child presenting with painful joints. • Know the key clinical features to assist in distinguishing the causes of a painful joint in children, in particular sepsis, reactive arthris, trauma and juvenile arthritis. • Know the key clinical findings of inflammatory arthropathies in children. • Outline the initial investigations in suspected juvenile arthritis. • Be able to outline the multidisciplinary managment plan that broadly addresses the following key area maintenance of joint function, analgesia & specific drug therapies, prevention and managment of disability.
4	18	• Know the key components of the history and examination in the child presenting with bruising. • Revise the coagulation pathway and in particular be able to correlate the steps of the pathway measured by prothrombin time, International normalised ration[INR] and activated partial thromboplasin time [aPTT] • Know the key clinical presentations of bleeding disorders due to vascular defects [eg Henoch-schonlein purpura], platelet disorders [eg idiopathic thrombocytopaenic purpura] and clotting factor deficiencies [eg haemophilia] • Know the typical features of idiopathic thrombocytopaenic purpura [ITP]. • Briefly outline the management options for ITP.
4	18	• Revise the key components of the immune system and be able to explain innate and adaptive immune responses. • Briefly outline the development of immune competency in children and explain the concepts of primary and secondary immunodeficiency. • Detail the key clinical features that suggest primary immunodeficiency.
4	18	• In the assessment of children presenting with fever, lymphadenopathy and oral mucosa changes, be able to outline the key features that support a diagnosis of vasculitis [Kawasaki disease]. • Know the key complications of Kawasaki disease and why early treatment with aspirin and immunoglobulins is required.
4	18	• Know the key clinical features that distinguish varicella-zoster [chicken-pox] infection from other vesicular skin rashes. • Know the natural history of V-Z infection. • List the complications of V-Z infection and describe the underlying pathophysiology. • Detail the management of V-Z infection, including the use of antiviral medication. • Outline the potential effects of V-Z infection of fetus in-utero. • Demonstrate an understanding of the current vaccination schedule for children by being able to outline a 'catch-up' vaccination program for a child who has missed vaccinations.
4	18	• Be able to appropriately assess and certify readiness to return to school or childcare for a child who has had an episode of communicable disease. • Be able to advise a parent on the recommended requirements for exclusion from and return to school for the following conditions: chickenpox, measles, whooping cough, hepatitis A, scabies.

CASE ONE

Short case number: 4_18_01
Category: Children & Young People
Musculoskeletal System & Skin
Discipline: Paediatrics_Medicine
Setting: General Practice
Topic: Child with knee pain & limp_juvenile arthritis

CASE

Jade Walton is 5 years old, she presents today with her mother. Jade has had a painful right knee for the last few days; her mother has noticed that Jade has been walking with a limp, especially when she gets up in the morning. Over the last few weeks she has had to stop gymnastics early because of pain in her knee. It does appear to be swollen compared to the left knee; as far as her mother knows she has not injured her knee.

QUESTIONS

1. What are the key components of your history in the assessment of Jade in particular what aspects of Jade's history will assist in distinguishing between sepsis, reactive arthritis and juvenile idiopathic arthritis?
2. What are the key components of your examination and why?
3. Your assessment of Jade reveals that she has had a sore knee for several weeks and that it is more painful in the morning. There is no recent history of viral infection or trauma. She is afebrile, with no rashes. The right knee is slightly swollen compared with the left and there are no signs of acute inflammation. You are concerned that Jade may have juvenile arthritis. What preliminary investigations would you undertake and why?
4. What extra-articular manifestations may be present in juvenile arthritis? What clinical features would you assess?
5. Preliminary investigations support a diagnosis of juvenile arthritis and you recommend assessment to a paediatrician. Briefly outline a management plan that address the following key aspects of management – maintenance of joint function, pain relief, specific drug therapy, prevention and management of disability.

Resources:

- **Murray KJ & Roberton DM. Arthritis and connective tissue disorders, Chapter 13 in Roberton DM, South M. Practical Paediatrics 6th Ed. Churchill Livingston, Edinburgh 2007. [pg 456-466]**
- **Munro JE, Piper SE. Limb and joint pain in children. Australian Doctor, How to treat. 26th March 2004. www.australiandoctor.com.au [accessed 29th May 2010]**

ANSWERS

Question 1
What are the key components of your history in the assessment of Jade in particular what aspects of Jade's history will assist in distinguishing between sepsis, reactive arthritis and juvenile idiopathic arthritis?

- Pain in or around the joint
- Warmth
- Swelling
- Occasionally erythema
- Loss of function, particularly loss of range of movement
- Stiffness, particularly early-morning.
- Fever
- Rash
- Family history of HLA B 27 associated disease
- History of antecedent infection
- Assess impact on function (sport, activities, sleep)

<u>Septic arthritis</u>
The hallmark of septic arthritis is a warm, tender joint with a dramatically restricted range of movement and an effusion. However, these signs are not always present. Septic arthritis of the shoulder or hip is very difficult to diagnose early, because of the lack of visible joint swelling in the early stages.

<u>Reactive arthritis</u>
Reactive arthritis conditions occur when, after (or during) a specific infection elsewhere in the body, a non-septic inflammation of joint(s) occurs for some period of time (usually less than 6 weeks). Transient synovitis of the hip or 'irritable hip' is considered a form of reactive arthritis that occurs in early childhood and is self-limiting. Post-streptococcal reactive arthritis and post-Mycoplasma reactive arthritis are increasingly recognized clinical entities, along with reactive arthritis induced by Salmonella, Yersinia and Campylobacter gastrointestinal infection. Usually, reactive arthritis has an oligoarthritis pattern, which may be severe in terms of pain, occasionally migrates, but lasts for a few weeks only.

<u>Juvenile idiopathic arthritis</u>
Inflammation in a joint for which no infective or other cause is found, and which persists for more than 6 weeks, is known as juvenile arthritis. The point of this definition is that other forms of arthritis (largely acute reactive arthritis) will have been diagnosed and/or resolved during this time

Question 2
What are the key components of your examination and why?

- Check for symptoms of fever , rash or stiffness (Reiter's arthritis or septic arthritis)
- Thorough examination of spine, sacroiliac joints, joints above and below the affected limb is essential (Juvenile idiopathic arthritis)
- Assess the gait
- Look for abductor weakness by the Trendelenburg test (slipped upper femoral epiphysis or Perthes disease
- Look for reduced internal rotation (Perthes disease)

- Measure leg length. A difference of >0.5cm is abnormal
- Examine the neurological system in particular looking for weakness

Question 3
Your assessment of Jade reveals that she has had a sore knee for several weeks and that it is more painful in the morning. There is no recent history of viral infection or trauma. She is afebrile, with no rashes. The right knee is slightly swollen compared with the left and there are no signs of acute inflammation. You are concerned that Jade may have juvenile arthritis. What preliminary investigations would you undertake and why?

The diagnosis of juvenile arthritis is made by excluding other conditions. It is a clinical diagnosis. Preliminary investigations may include

- X-ray of the knee to exclude bony pathology
- FBC and ESR and CRP to exclude possibility of septic joint
  - Note leukaemia is the great mimicker of joint pain and a normal FBC with a child who appears unwell should not reassure the doctor. Further investigations in an unwell child are always warranted.

Question 4
What extra-articular manifestations may be present in juvenile arthritis? What clinical features would you assess?

- Eye Disease
  - Chronic anterior uveitis which may lead to permanent scaring and blindness.
  - Glaucoma
  - Cataracts
- Cardiac complications including pericarditis, myocarditis, heart failure
- Growth Retardation secondary to long term corticosteroid use.

Question 5
Preliminary investigations support a diagnosis of juvenile arthritis and you recommend assessment to a paediatrician. Briefly outline a management plan that address the following key aspects of management – maintenance of joint function, pain relief, specific drug therapy, prevention and management of disability.

- Pain relief
  - NSAIDS (piroxicam is TGA approved for children)
- Joint aspiration and steroid injection
  - For younger children when a single joint is involved
- DMARDS
  - 3rd line
  - Eg methotrexate
- Physiotherapy
  - Help with joint positioning and prevention of muscle weakness
- Occupational Therapy
  - For help with activities of daily living
- Regular review with ophthalmologist to assess for extra ocular involvement

CASE TWO

Short case number: 4_18_02
Category: Children & Young People.
Immunological and haematological systems.
Discipline: Paediatrics_Medicine
Setting: Emergency Department
Topic: Bruising & bleeding_idiopathic thrombocytopaenic purpura.

CASE

Joseph Teo is 3 years old, he presents to the emergency department with his parents who are very concerned about the bruises on his legs. They have noticed them appearing over the last few 24 hours. Joseph has been unwell over the last few days with a cough, runny nose and mild temperature, but he has been eating and drinking well and still appears to be fairly active.

On examination he is alert and does not look unwell.
afebrile, BP 100/60; HR RR
No pallor. He has multiple obvious bruises over both legs and on his arms

QUESTIONS

1. What are the key differential diagnoses that you would consider and how what history and examination would you undertake to assist you in reaching a diagnosis?
2. You know that it is possible that Joseph may have a bleeding disorder and the Paediatric registrar asks you to order coagulation tests. Revise the coagulation pathway, in particular the components that are measured by International normalised ratio [INR] and activated partial thromboplasin time [aPTT]
3. Summarise the key clinical features of bleeding disorders under the following headings and provide an example of each condition, vascular defects, platelet disorders and clotting factor deficiencies.
4. Joseph's Full blood count reveals a platelet count of 80 x 10^6/L; other parameters were in the normal range. The Paediatrician explains to Joseph's parents that he idiopathic thrombocytopaenic purpura [ITP]. Explain the key clinical features of ITP.
5. Briefly outline the management of ITP, explaining the options of first line and second line therapy.

Resources:

- **Saxon B. Abnormal bleeding and clotting. Chapter 16 in Roberton DM, South M. Practical Paediatrics 6th Ed. Churchill Livingston, Edinburgh 2007. [pg 560-569]**

ANSWERS

Question 1
What are the key differential diagnoses that you would consider and how what history and examination would you undertake to assist you in reaching a diagnosis?

<u>Differential Diagnosis:</u>

- immune thrombocytopenia purpura
- aplastic anaemia

<u>History:</u>

1. Is the bleeding abnormal?

The main question to answer in the history is whether the bleeding symptoms are within or outside normal limits. Isolated bruises over the shins are common, while spontaneous petechiae are abnormal. Finger-induced epistaxis is common and not indicative of a bleeding disorder; however, recurrent nose bleeds lasting more than 10 minutes or leading to anaemia are often related to a bleeding disorder. Table 16.2.1 gives some clinical guidance

2. When did the bleeding start (Prenatal and neonatal)

- congenital infection may result in a bleeding disorder
- mucosal bleeding occurs with haemorrhagic disease of the newborn
- umbilical stump bleeding is associated with factor XIII deficiency and dysfibrinogenaemias
- intracranial haemorrhage may occur with factor deficiencies and with neonatal alloimmune thrombocytopenia
- prolonged bleeding following circumcision is suggestive of haemophilia and may be the presenting feature of haemorrhagic disease of the newborn

3. Early childhood bleeding

- often implies a congenital defect
- bruising, muscle and joint bleeding is strongly suggestive of haemophilia
- petechiae and mucosal bleeding suggests a platelet problem or von Willebrand disorder

4. Sudden onset bleeding

- usually an acute problem such as immune thrombocytopenic purpura
- Non-accidental injury may have a haemorrhagic presentation with inadequate explanations for each specific bruise, which may have an unusual distribution (Ch. 3.9). Skeletal trauma and other stigmata of non-accidental injury may be present

5. Where is the bleeding?

- Joint bleeding: haemophilia A and B

- Nasal mucosa: local irritation; von Willebrand disorder and platelet dysfunction
- Gums, periosteum, skin: scurvy
- Gastrointestinal: haemorrhagic disease of the newborn in babies; liver disease in older children
- Retro-orbital: haematological malignancy or disseminated solid tumour.

6. Family History

Haemophilia A and B are X-linked; most von Willebrand disorder subtypes and haemorrhagic hereditary telangiectasia are recessive and several platelet function disorders are dominantly inherited. Clinical penetrance in haemophilia carriers and von Willebrand disorder may be variable.

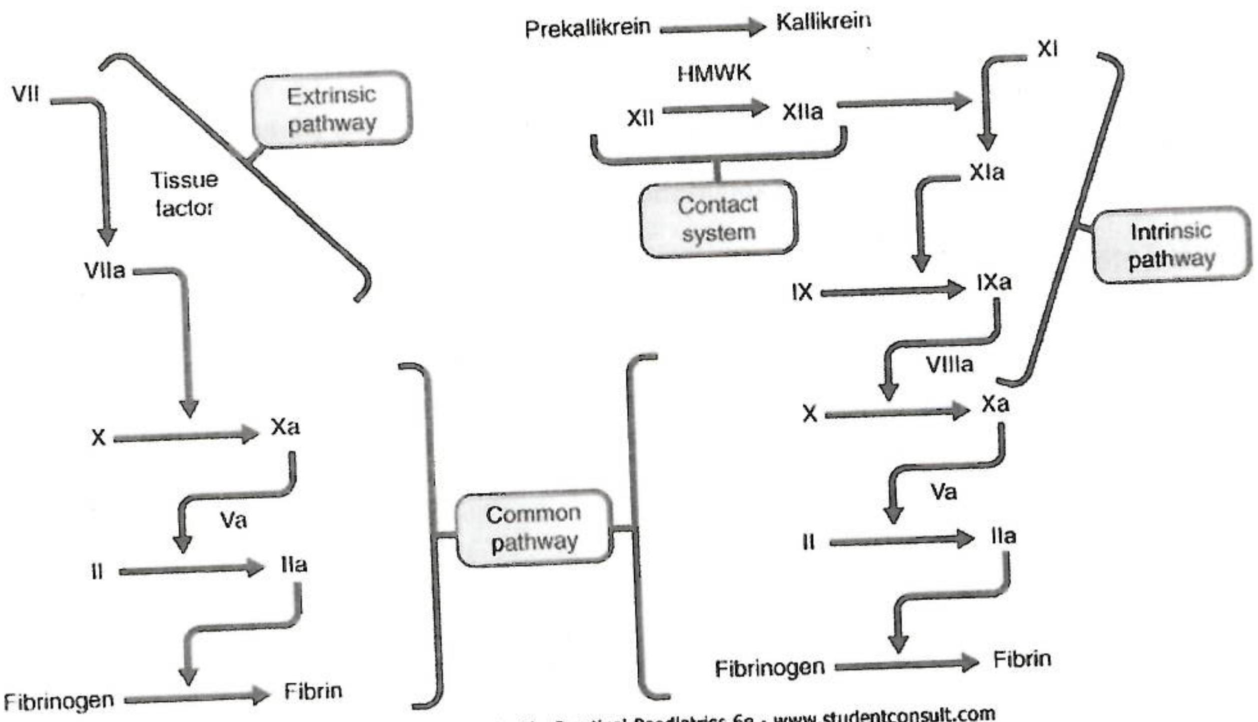
7. Past history of bleeding

Easy bruising, bruising at abnormal sites, prolonged bleeding following trivial trauma or bleeding following surgery and dental extractions are all indications for investigation.

Question 2
You know that it is possible that Joseph may have a bleeding disorder and the Paediatric registrar asks you to order coagulation tests. Revise the coagulation pathway, in particular the components that are measured by International normalised ratio [INR] and activated partial thromboplastin time [aPTT]

<u>Coagulation pathway</u>

- The aPTT and PT/INR are screening tests which represent how various 'factors' interact in the test tube. Useful tests to determine which factor may be low
- In vivo coagulation does not occur as a 'cascade', rather a series of positive feedback reactions starting with tissue factor and factor VII and culminating in a thrombin burst
- Clotting occurs at the site of injury by thrombin cleaving fibrinogen to fibrin
- Thrombin inhibits coagulation by binding with thrombomodulin on normal blood vessels



The coagulation system as measured by initial blood tests: prothrombin time (INR; left) and aPTT (right). HMWK, high-molecular-weight kininogen; 'a' indicates activated factor. This figure does not represent in vivo coagulation, rather the coagulation factors (in test tubes) that influence the prothrombin time and aPTT.

Question 3
Summarise the key clinical features of bleeding disorders under the following headings and provide an example of each condition, vascular defects, platelet disorders and clotting factor deficiencies

<u>Vascular defects</u>
Examples include anaphylactoid purpura (Henoch-Schönlein purpura), infective states, and nutritional deficiency. Henoch-Schonlein Purpura is readily recognized by the characteristic distribution of the rash over the buttocks, legs and backs of the elbows. Frequently, it is accompanied by abdominal pain, melaena, joint swellings and occasionally a glomerulonephritis

<u>Platelet disorders</u>
Bleeding disorders resulting from platelet abnormalities are usually due to thrombocytopenia but may be due to qualitative platelet defects
Eg acute immune thrombocytopenia (clinical features listed above)

<u>Clotting factor deficiencies</u>
Examples include: Haemophilia A (factor VIII deficiency): and Haemophilia B (Christmas disease, factor IX deficiency): Characteristically skin and soft tissue bleeds are common in the first year and beyond. Haemarthroses usually only occur once the child is walking. The ankles are common bleeding sites in young children; elbow and knee bleeding occur more commonly in older children

Question 4

Joseph's Full blood count reveals a platelet count of 80 x 10^6/L; other parameters were in the normal range. The Paediatrician explains to Joseph's parents that he idiopathic thrombocytopaenia purpura [ITP]. Explain the key clinical features of ITP.

Classical features of immune thrombocytopenia purpura

- 80-90% of paediatric immune thrombocytopenic purpura cases
- preceding viral illness is common
- peak age 2-5 years
- abrupt onset of bleeding
- mucosal and skin bleeding
- petechiae common
- otherwise normal examination, i.e. no lymphadenopathy or hepatosplenomegaly
- platelet count usually <20 × 10^9 /L
- normal red cell and white cell parameters

Question 5

Briefly outline the management of ITP, explaining the options of first line and second line therapy.

Treatment options are considered if there is problematic bleeding or if the platelet count is <10 x 10^9/L

1st line options

- Conservative. - No drug treatment. Recovery of platelets may take up to 4 – 6 months. <1% risk ICH.
- Low dose corticosteroids
- High dose corticosteroids (recovery <1 week)
- Intravenous gammaglobulin

2nd line options

- Anti RHD antibody
- Splenectomy

CASE THREE

Short case number: 4_18_03
Category: Children & Young People
Immume & Haematology systems.
Discipline: Paediatrics_Medicine
Setting: General Practice
Topic: Recurrent Infections_immune system.

CASE

Jay is 8 months old, he has a 3 day history of cough, rhinorrhoea and fevers to 39^{0}C. Today his symptoms have improved and his fever has settled but he has developed this rash. On examination you note a diffuse morbilliform rash on Jay's back. Your diagnosis is roseola infantum. As you explain to Jay's mum that the cause is a viral infection, she sighs, *"He seems to have so many infections, and he is only 7 months old....is this normal? I am worried he may have something wrong with his immune system"*

QUESTIONS

1. In your assessment of Jay, what key features of the history and examination would suggest primary immunodeficiency?
2. Your assessment of Jay does not detect any concerning features and you explain to Jay's mother that viral infections are quite common in young children. Summarise the normal development of immune competency in children and explain the innate and adaptive immune responses.
3. Following your explanation about the immune system and antibody development, Jay's mother seems confused...she asks what happens when Jay is vaccinated and if this is different to a viral infection, what would you explain to her?

Resources:

- **Wong M, Immunodeficiency and its investigation, Chapter 13 Isaacs D, in Roberton DM, South M. Practical Paediatrics 6th Ed. Churchill Livingston, Edinburgh 2007. [pg 443-455]**
- **Myths and Realities. Responding to arguments against immunisation. A guide to providers. Australian Government Department of Health and Ageing. 4th Edition 2008**

ANSWERS

Question 1
In your assessment of Jay, what key features of the history and examination would suggest primary immunodeficiency?

Deficiencies of humoral immunity present in the second half of the first year of life, when maternally derived antibody has waned. Significant deficiencies of T-cell function present within the first few months of life. Many primary immunodeficiencies present in infancy with dermatological manifestations such as severe or atypical eczema, thrombocytopenic purpura, recalcitrant candidiasis and abscesses, or are diagnosed in association with other conditions such as cardiac, endocrine and neurological anomalies, such as DiGeorge syndrome and ataxia telangiectasia.

Question 2
Your assessment of Jay does not detect any concerning features and you explain to Jay's mother that viral infections are quite common in young children. Summarise the normal development of immune competency in children and explain the innate and adaptive immune responses

Immune defence is provided by multiple components, which can be categorized as two main groups

- Innate, non-antigen-specific mechanisms:
  - barriers: epithelial surfaces, mucosal barriers
  - secretions: saliva, respiratory secretions, tears, urine
  - normal microbial flora: gastrointestinal, genital tract
  - phagocytic cells: neutrophils, macrophages
  - natural killer cells
  - proteins: complement, mannose-binding lectin, antimicrobial peptides
- Adaptive, antigen-specific immune responses: these are the basis of immunological memory and are essential for maturation of protective immune responses and efficacy of vaccination. The components are:
  - T cells
  - B cells and antibody

The adaptive immune response develops with recurrent exposure to infection. Primary exposure often results in clinically evident infection and occurs most frequently in infancy and early childhood. Secondary exposure in the presence of an intact adaptive immune system results in a more rapid and efficient response, and avoidance of subsequent infection and clinical symptoms in older children and adults. In association with increasing exposure, this results in the peak number of infections between the ages of 2 and 4 years of life, with an average of six infections a year. Existence of siblings, day care attendance and exposure to cigarette smoke further increases this number.

Question 3
Following your explanation about the immune system and antibody development, Jay's mother seems confused...she asks what happens when Jay is vaccinated and if this is different to a viral infection, what would you explain to her?

Some worry that vaccines weaken the immune system especially when given to babies. Vaccines do not weaken the immune system; rather they strengthen it by inducing protection against specific disease.

Children are exposed every day to many foreign antigens through routine activities such as eating, drinking and playing. Therefore childrens immune systems become very robust. Vaccines only contain a small number of antigens compared to what children are exposed to every day and therefore do not 'use' up or 'exhaust the immune system.

CASE FOUR

Short case number: 4_18_04
Category: Children & Young People
Immune & haematological systems.
Discipline: Paediatrics_Medicine
Setting: Paediatric Ward
Topic: Vasculitis Syndromes_Kawasaki disease

CASE

You are the Paediatrics' intern; you receive a call from the registrar explaining that Emily Brenton who is 4 years old is to be admitted. She has been referred by her GP because she has been unwell for 5 days with high fevers, poor fluid intake due mouth ulcers and today has developed a generalised rash and appears worse. The registrar is concerned that she may have Kawasaki disease.

QUESTIONS

1. You have never heard of Kawasaki disease so while waiting for Emily to arrive on the ward you quickly find a computer and look it up. What are the key clinical features that you would assess when Emily arrives on the ward?
2. What are the diagnostic criteria for Kawasaki disease and what is the key differential diagnosis?
3. Emily arrives on the ward, with her mother. She is listless and unwell and her mother is very concerned about the diagnosis and wants to know why they had to come to hospital. Explain to Emily's mother major complications of Kawasaki disease and the treatment that is required.
4. What are possible long term consequences of Kawasaki disease?

Resources:

- **Murray KJ & Roberton DM. Arthritis and connective tissue disorders, Chapter 13 in Roberton DM, South M. Practical Paediatrics 6th Ed. Churchill Livingston, Edinburgh 2007. [pg 470-471]**

ANSWERS

Question 1
You have never heard of Kawasaki disease so while waiting for Emily to arrive on the ward you quickly find a computer and look it up. What are the key clinical features that you would assess when Emily arrives on the ward?

- fever
- conjunctival injection (non purulent)
- oral mucosal changes
- swelling of hands, feet, erythema of palms or soles, or peeling of skin
- tender right upper quadrant (cholecystitis)
- swollen joints (arthritis)
- morbilliform rash

Question 2
What are the diagnostic criteria for Kawasaki disease and what is the key differential diagnosis?

Diagnostic or classification criteria have been developed. They are a fever of more than 38.5°C for 5 or more days in the absence of evidence of a streptococcal (or other specific) infection with at least four of the following five manifestations

- bilateral non-purulent conjunctival injection
- oral mucosal changes (erythema or dry cracked lips or strawberry tongue)
- cervical lymphadenopathy with one node 1.5 cm or more in diameter
- changes in the extremities (swelling of hands or feet, or erythema of palms or soles, or membrane-like peeling of the skin)
- a generalized rash, which often is morbilliform in appearance.

Question 3
Emily arrives on the ward, with her mother. She is listless and unwell and her mother is very concerned about the diagnosis and wants to know why they had to come to hospital. Explain to Emily's mother major complications of Kawasaki disease and the treatment that is required.

If left untreated Kawasaki disease can cause coronary arteritis which if left untreated can cause coronary aneurysms leading to heart disease.

Question 4
What are possible long term consequences of Kawasaki disease?

A major complication of this disorder, reflecting its vasculitic nature, is a propensity to develop proximal arterial aneurysms. This is seen in the coronary arteries particularly and is found in approximately 20% of untreated patients on echocardiography. The incidence of coronary artery aneurysm formation can be reduced greatly by treatment with intravenous immunoglobulin (IVIG) during the early phase of the acute illness. Steroids may play a role in those who are resistant to IVIG. Aspirin is used in the early phase to prevent thrombosis. Atherosclerosis and ischaemic heart disease may follow later in life.

CASE FIVE

Short case number: 4_18_05
Category: Children & Young People
Infectious Diseases
Discipline: Paediatrics_Medicine
Setting: General Practice
Topic: Varicella- Zoster infection.

CASE

Alannah who is 34 weeks pregnant with her third child presents with her second child Rodney, who is 3 years old. Rodney has been unwell for a few days with a sore throat and temperature to 38^{0}C. Yesterday Alannah noticed some spots on his forehead that she thought were insect bites, however today there are a lot more spots all over his face.
You notice that some of the lesions are vesicular and your provisional diagnosis is varicella [chicken-pox].
Rodney missed his 18 month and 2 year old vaccinations because they were overseas at the time.

QUESTIONS

1. What further history would you explore and why?
2. What clinical features distinguish varicella from other cause of vesicular rashes?
3. You confirm your provisional diagnosis of chicken-pox and you explain this to Alannah. What would you explain to Alannah and what are the possible complications that she should be alert to? Outline your management plan for Rodney.
4. Alannah asks about Harry her 5 year old son; *"Should I keep him home from Kindergarten, He has had all his vaccinations including chickenpox at 18months?"* What would you explain to Alannah?
5. As you are explaining about how infectious chicken pox is Alannah looks very worried, *"What about me, I have never had chicken pox and I am 34 weeks pregnant, could I get chicken-pox and pass it onto the baby?"* What would you explain to Alannah? Outline your management plan for Alannah.
6. What 'catch-up' plan would you advise for Rodney's immunisation once he recovers from this illness?

Resources:

- **Isaacs D. Infections in Childhood. Chapter 12 in Roberton DM, South M. Practical Paediatrics 6th Ed. Churchill Livingston, Edinburgh 2007. [pg 389-390]**

Electronic resources:

- **Litt J, BurgessM. Varicella and Varicella Vaccination. Australian Family Physician. Vol 32; No8. August 2003. [accessed online 10th July 2010]**
- **McKinnon, Howard, Evaluating the febrile Patient with Rash, American Family Physician Aug 15 2000 [accessed online 28th September 2010]: http://www.aafp.org/afp/20000815/804.html**

ANSWERS

Question 1
What further history would you explore and why?

- Alannah
  - Antenatal history including complications, current varicella serology.
  - Current condition
  - Medications
  - Other children? vaccinated
  - Social – family support
- Rodney
  - Immunisation history
  - PMHx
  - Current symptoms (check for complications of varicella including pneumonia and encephalitis)
- Explore underlying reasons for missed vaccinations. Suggest catch up doses.

Question 2
What clinical features distinguish varicella from other cause of vesicular rashes?

- Herpes Zoster will present with a prodrome of unusual skin sensations; dermatomal pattern, with lesions rarely crossing midline; pain often severe; more commonly in thoracic and facial dermatomes.
- Staphylococcal bacteremia may present with a widespread pustular eruption. Gonococcemia may also produce a pustular rash, although other lesion types, such as macules, petechiae and papules, are usually present.

Question 3
You confirm your provisional diagnosis of chicken-pox and you explain this to Alannah. What would you explain to Alannah and what are the possible complications that she should be alert to? Outline your management plan for Rodney.

- Bacterial superinfection of skin
- Varicella pneumonitis occurs in immunocompromised children but also in pregnant women and normal adults.
- Bacterial pneumonia (pneumococcal or staphylococcal) can rarely complicate varicella pneumonitis.
- Encephalitis: (incidence about 1 in 1000 cases) The most common form is pure cerebellar ataxia, with complete recovery over days to weeks. The severe form is acute disseminated encephalomyelitis (ADEM), a postinfectious demyelinating illness with high morbidity and mortality

In immunocompetent children no specific therapy is indicated. Symptomatic treatment consists of Calamine lotion, cool compresses, and possibly oral antihistamines at night to improve sleep. Keeping the skin cool may reduce the number of lesions. Scratching increases the risk of secondary bacterial infection - cut the child's nails short at the first sign of the disease. Avoid aspirin.

The patient is infectious from one to two days before the onset of the rash until the lesions have fully crusted over. Children must be excluded from school until fully recovered (all lesions crusted over) or at least one week after the eruption first appears.

Question 4
Alannah asks about Harry her 5 year old son; "Should I keep him home from Kindergarten, He has had all his vaccinations including chickenpox at 18months?" What would you explain to Alannah?

Seroconversion following vaccination occurs in 90- 100% of individuals. 70- 90% of vacinees are protected when exposed, and vacinees who develop infection after being exposed usually have mild disease. The duration of immunity after vaccination is thought to be substantial but it is not yet established and it could be that a booster dose is necessary.

You would advise Alannah that it is most likely that Harry has immunity but it cannot be guaranteed. It may be advisable to keep Harry away from Kindergarten until the incubation period is over (Incubation period 10-21 days – the usual is 14 and a short prodromal period of 1-2 days before the rash appears)

Question 5
As you are explaining about how infectious chicken pox is Alannah looks very worried, "What about me, I have never had chicken pox and I am 34 weeks pregnant, could I get chicken-pox and pass it onto the baby?" What would you explain to Alannah? Outline your management plan for Alannah

The implications of primary VZV infection in pregnancy for the mother and for the fetus vary with the period of gestation. For the mother, the risk of adverse effects is greatest in the third trimester, whereas for the fetus the risk is greatest in the first and second trimesters

Seropositive mothers do not need any treatment. A blood test should be done immediately in Alannah to determine her immunity.

In seronegative women Zoster immunoglobulin (ZIG) should be offered if there is a history of significant exposure to varicella-zoster virus (VZV) (chickenpox) infection.

If siblings at home have chickenpox, a newborn baby should be given ZIG if its mother is seronegative. If the mother is seropositive the newborn baby does not need to be isolated from siblings.

Question 6
What 'catch-up' plan would you advise for Rodney's immunisation once he recovers from this illness?

Rodney does not need the 18th month varicella vaccination as he has developed immunity from contracting the disease, however he should have the routine 2 year old vaccinations as soon as possible.